

Hybrid Universal Control in a 2x2 Qubit+Cavity Logical Subspace: Selective vs Native Gate Families in

`cqed_sim`

Codex

cQED Autonomous Research Platform

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Abstract

We study a practical control-design question for circuit quantum electrodynamics: which gate family is the best realistic route to hybrid universal control of a qubit and cavity when the logical task is restricted to the smallest nontrivial 2×2 block? The benchmark is implemented on top of `cqed_sim` for a dispersive transmon-storage system with $\omega_q/2\pi = 6.150$ GHz, $\omega_c/2\pi = 5.241$ GHz, $\alpha/2\pi = -255$ MHz, $\chi/2\pi = -2.84$ MHz, $\chi'/2\pi = -21$ kHz, and $K/2\pi = -28$ kHz. We compare six routes: (A) baseline dispersive control with qubit rotations, cavity displacements, and SNAP; (B) selective hybrid control with SQR-like primitives; (C) ECD-like conditional displacement; (D) a minimal native chi-wait entangler library; (E) waveform-level GRAPE; and (F) native SWAP-/sideband-style control using Jaynes-Cummings and blue-sideband exchange primitives. During the study, the local `cqed_sim` copy was extended so that conditional displacement and exchange-style native interactions are available as first-class synthesis primitives rather than study-local wrappers.

The main result is that the best practical route is still a hybrid answer, but the native-gate comparison is now substantially sharper. The exact logical entangler is obtained most efficiently by native dispersive waiting plus fast qubit rotations: $F_{\text{strict}} = 0.9999999$ in 256 ns with negligible leakage. The best local cavity action still comes from the older dispersive D -SNAP- D route, which reaches $F_{\text{strict}} = 0.9887$ with 1.85% average leakage. The new SWAP-/sideband-style native family is a serious fast benchmark: it reaches $F_{\text{strict}} = 0.8784$ for the local task in 440 ns and improves on the selective SQR library, but it remains leakier and less exact than the SNAP local gate and the native chi-wait entangler. GRAPE is useful as a pulse benchmark, yet it does not beat the native entangler and only challenges the local gate on speed, not fidelity. For this smallest Fock-encoded hybrid block, the recommended route is therefore an augmented dispersive library $\{R_q, D, \text{SNAP}, \text{native } \chi\text{-wait}\}$, with native exchange primitives retained as a promising path for larger or less Fock-like encodings.

1 Introduction

Circuit QED offers a natural hybrid architecture in which a nonlinear ancilla qubit controls a high-coherence bosonic mode [1, 2]. The resulting control question is no longer just whether a gate set is universal in principle, but whether it is the best practical choice once gate times, selectivity, leakage, calibration effort, and decoherence are all taken seriously. That distinction is especially important for bosonic and hybrid control because several paradigms compete on different operating principles.

The most mature dispersive route combines cavity displacements with photon-number selective phase gates (SNAP) and ancilla rotations [3–5]. Faster echoed conditional displacement (ECD) strategies can outperform selective control in weak-dispersive regimes [6], and more recent conditional-displacement variants such as CNOD push the same idea toward fast structured

multimode control [7]. A different native route is to activate exchange or sideband interactions directly. Sideband control has been part of the circuit-QED toolkit since the early qubit-photon entanglement experiments of Wallraff *et al.* [8], and more recent work shows that one can rapidly transpose between dispersive and resonant-like light-matter regimes [9] or use sideband-enhanced SWAP control in weakly coupled multimode bosonic memories [10]. Finally, native activated nonlinear interactions show what bosonic control can look like when the hardware itself supplies stronger direct nonlinearities than a dispersive ancilla can offer [11].

This study asks which of these ideas is best suited to the local simulator and device assumptions already encoded in `cqed_sim`. To keep the question sharp, we start from the smallest hybrid logical space, $\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$, and compare gate libraries for two tasks: a local cavity basis change and a maximally entangling hybrid unitary. Relative to the initial version of the study, the present revision makes two important changes. First, native-gate families are compared explicitly against selective ones. Second, the local `cqed_sim` copy is extended so ECD-like conditional displacement and SWAP-/sideband-style exchange interactions can be benchmarked through the synthesis stack rather than only discussed conceptually. The remainder of the paper presents the model and benchmark, the candidate libraries, the comparative results, the validation checks, and the resulting recommendation.

2 System and Methods

2.1 Hamiltonian

The static model is the dispersive transmon–cavity Hamiltonian supported by `cqed_sim`,

$$\frac{H_0}{\hbar} = \omega_c a^\dagger a + \omega_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2 + \chi a^\dagger a b^\dagger b + \frac{K}{2} a^{\dagger 2} a^2 + H_{\chi'}, \quad (1)$$

where a is the cavity annihilation operator, b is the transmon lowering operator, α is the transmon anharmonicity, χ is the leading dispersive shift, K is the cavity self-Kerr term, and $H_{\chi'}$ denotes the first higher-order dispersive correction encoded in the model through `chi_higher=( $\chi'$ ,)`. The direct-control benchmark uses the drive Hamiltonian

$$\frac{H_d(t)}{\hbar} = \epsilon_c(t) a + \epsilon_c^*(t) a^\dagger + \epsilon_q(t) b + \epsilon_q^*(t) b^\dagger, \quad (2)$$

with independently addressable cavity and qubit I/Q controls.

The native exchange benchmark is motivated by the effective red/blue-sideband and Jaynes–Cummings interaction forms already supported by the simulator:

$$\frac{H_{JC}}{\hbar} = g_{JC} \left(a^\dagger \sigma_- e^{-i\phi_{JC}} + a \sigma_+ e^{i\phi_{JC}} \right), \quad (3)$$

$$\frac{H_{BS}}{\hbar} = g_{BS} \left(a^\dagger \sigma_+ e^{-i\phi_{BS}} + a \sigma_- e^{i\phi_{BS}} \right), \quad (4)$$

which represent ideal rotating-wave effective interactions rather than a full calibrated pulse model. These are used at the synthesis layer as `JaynesCummingsExchange` and `BlueSidebandExchange`.

The logical basis is the four-dimensional block

$$\mathcal{H}_{\text{logical}} = \text{span}\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}. \quad (5)$$

Two benchmark targets are used:

$$U_H = I_q \otimes H_c, \quad (6)$$

where H_c is the Hadamard on the cavity Fock pair $\{|0\rangle, |1\rangle\}$, and the cavity-controlled qubit flip

$$U_{CX} = |0\rangle\langle 0|_c \otimes I_q + |1\rangle\langle 1|_c \otimes X_q. \quad (7)$$

2.2 Simulation Parameters

Table 1: Nominal system and control parameters used in the expanded benchmark.

Parameter	Value	Unit
Qubit frequency $\omega_q/2\pi$	6.150	GHz
Cavity frequency $\omega_c/2\pi$	5.241	GHz
Anharmonicity $\alpha/2\pi$	-255	MHz
Dispersive shift $\chi/2\pi$	-2.84	MHz
Higher-order dispersive shift $\chi'/2\pi$	-21	kHz
Cavity self-Kerr $K/2\pi$	-28	kHz
Cavity truncation n_{cav}	8	Fock states
Transmon truncation n_{tr}	2	levels
Qubit rotation duration	40	ns
Displacement duration	80	ns
Selective gate duration	1100	ns
Conditional-displacement prior	160	ns
Jaynes–Cummings exchange prior	120	ns
Blue-sideband exchange prior	120	ns
Native dispersive wait $\pi/ \chi $	176	ns
GRAPE amplitude bound $ u_j /2\pi$	40	MHz
Qubit T_1	30	μs
Qubit T_2	20	μs
Cavity T_1	250	μs

2.3 Computational Approach

The study uses three layers of `cqed_sim` functionality [12]. First, the device model is instantiated with `DispersiveTransmonCavityModel` and a rotating `FrameSpec`. Second, the decomposition study uses `Subspace`, `UnitarySynthesizer`, `QubitRotation`, `Displacement`, `SNAP`, `SQR`, `FreeEvolveCondPhase`, `ConditionalDisplacement`, `JaynesCummingsExchange`, and `BlueSidebandExchange`. Third, the pulse benchmark uses `build_control_problem_from_model`, `ModelControlChannelSpec`, `PiecewiseConstantTimeGrid`, `UnitaryObjective`, and `GrapeSolver`.

During this revision, the local `cqed_sim` copy was extended so that `ConditionalDisplacement`, `JaynesCummingsExchange`, and `BlueSidebandExchange` are first-class sequence primitives in `cqed_sim.unitary_synthesis`. They were also added to the fast-eval path, exported through the package `__init__`, documented in the API reference, and covered by dedicated unit tests. The remaining gap is not operator availability but calibrated pulse export: the native primitives are effective-interaction models, not yet waveform-bridge-backed pulse recipes.

Six libraries are compared:

$$\mathcal{G}_A = \{R_q, D, \text{SNAP}\} \text{ with native } \chi\text{-wait allowed}, \quad (8)$$

$$\mathcal{G}_B = \{R_q, D, \text{SQR}\}, \quad (9)$$

$$\mathcal{G}_C = \{R_q, D, \text{CD}\}, \quad (10)$$

$$\mathcal{G}_D = \{R_q, D, U_{\chi\text{-wait}}\}, \quad (11)$$

$$\mathcal{G}_E = \{\text{piecewise qubit and cavity I/Q drives}\}, \quad (12)$$

$$\mathcal{G}_F = \{R_q, U_{\text{JC}}, U_{\text{BS}}\}. \quad (13)$$

Here CD is the first-class conditional-displacement primitive, and U_{JC} and U_{BS} are the Jaynes–Cummings and blue-sideband exchange primitives used to represent SWAP-/sideband-style na-

tive control. Library D isolates the native dispersive entangler already present in the hardware model.

For every candidate we report strict logical fidelity,

$$F_{\text{strict}} = \frac{1}{d^2} \left| \text{Tr} \left(U_{\text{target}}^\dagger U_{\text{logical}} \right) \right|^2, \quad (14)$$

with $d = 4$, the average leakage outside the logical block, total duration, and a composite score

$$J = F_{\text{logical}} - 0.20 L - 0.02 \frac{T}{1000 \text{ ns}}. \quad (15)$$

We also report a block-gauge fidelity, useful when a candidate reproduces the correct entangling structure up to a removable cavity-block phase.

3 Results

3.1 Goal 1: Local cavity action on the 2x2 block

Figure 1 and Table 2 summarize the local cavity benchmark. The best exact local action is still not the fastest pulse, but the older dispersive D -SNAP- D route. Gate Set A reaches $F_{\text{strict}} = 0.9887$ with $L = 0.0185$ in 1260 ns. This remains the best strict and block-gauge score in the study and stays extremely stable under perturbation replay and truncation checks.

The GRAPE benchmark is competitive but does not win the fidelity-first metric on this target. The best nominal GRAPE local case is $E_{\text{local},320}$, which reaches $F_{\text{strict}} = 0.9618$ in 320 ns with $L = 0.0524$. This is a meaningful speed advantage, but it does not close the accuracy gap to the structured SNAP-based decomposition.

The most important new local result is that the native SWAP-/sideband-style family is a real intermediate candidate rather than a thought experiment. Gate Set F reaches $F_{\text{strict}} = 0.8784$ with $L = 0.0788$ in 440 ns. It therefore beats the selective SQR-based local library (Gate Set B, $F_{\text{strict}} = 0.8615$, 2440 ns) on both fidelity and speed, but still falls short of the SNAP baseline and of the best GRAPE pulse. The ECD-like conditional-displacement library and the minimal native local probe from Gate Set D both perform poorly for the strict Fock encoding, indicating that neither simple displacement geometry nor bare chi-waiting alone is a good local cavity-control primitive for this logical block.

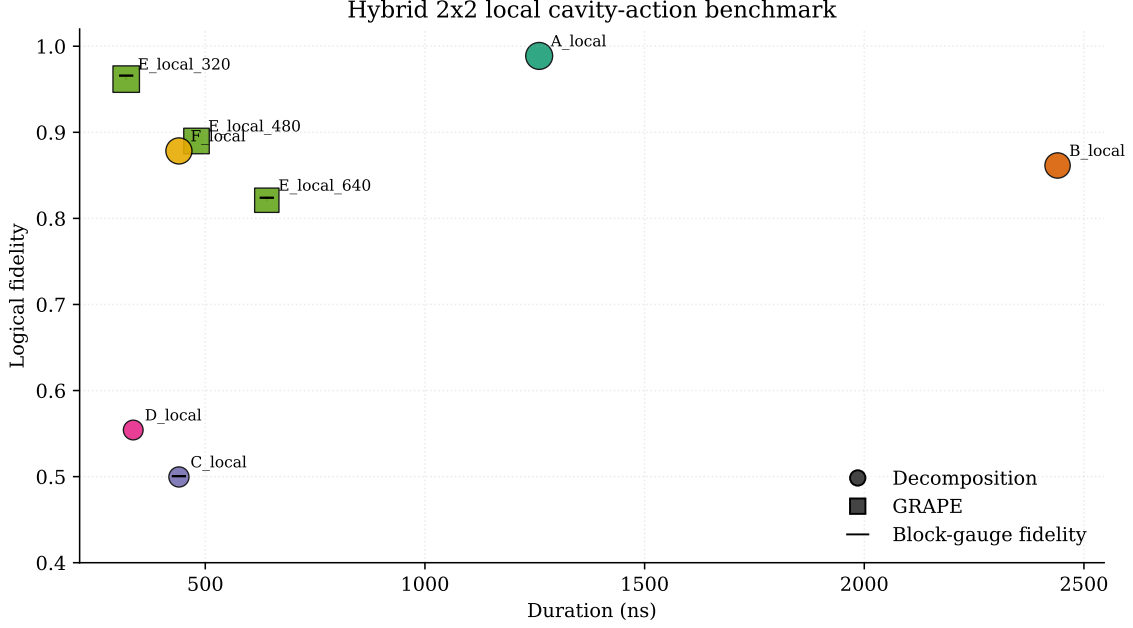


Figure 1: Local cavity-action benchmark for the target $I_q \otimes H_c$. Point position gives duration and strict logical fidelity. Marker size decreases with leakage. The short black tick above a point indicates the block-gauge fidelity when it differs from the strict fidelity.

Table 2: Best local-control candidates for the cavity Hadamard target.

Library	Duration (ns)	F_{strict}	Leakage	Score J
A: D -SNAP- D	1260	0.9887	0.0185	0.9598
E: GRAPE (320 ns)	320	0.9618	0.0524	0.9449
F: native SWAP/sideband	440	0.8784	0.0788	0.8538
B: selective hybrid	2440	0.8615	0.1350	0.7857
D: native local probe	336	0.5542	0.4747	0.4525
C: ECD-like CD	440	0.4996	0.4535	0.4001

3.2 Goal 2: Hybrid entangling unitary synthesis

The entangler benchmark produces an even clearer answer. The native dispersive conditional phase combined with two fast qubit rotations implements the target $CX(c \rightarrow q)$ essentially exactly. Gate Set D reaches $F_{\text{strict}} = 0.9999999$ and $L \approx 0$ in 256 ns, and the same sequence is recovered inside Gate Set A because the actual resource is the native chi-wait primitive rather than SNAP itself. This is the dominant result of the study: for the 2x2 Fock benchmark, the most important hybrid primitive is already present in the native Hamiltonian and does not require either a selective hybrid construction or a waveform-optimized pulse to become useful.

GRAPE again provides a useful reference but not a better answer. The best nominal GRAPE entangler is $E_{\text{ent},400}$, which reaches $F_{\text{strict}} = 0.9458$ in 400 ns with $L = 0.0854$. That is a respectable benchmark, but it is neither faster nor cleaner than the native dispersive entangler.

The expanded native comparison clarifies what SWAP-/sideband-style exchange can and cannot do in this smallest logical block. Gate Set F reaches $F_{\text{strict}} = 0.7510$ with $L = 0.1834$ in 320 ns. This is substantially faster and slightly more faithful than the selective SQR entangler on the strict metric, but it still falls well short of the exact native conditional-phase route. Gate Set B remains physically interesting because a single SQR entangler reaches block-gauge

fidelity 1.0 but strict fidelity $1/\sqrt{2}$, which means the entangling structure is exactly right up to a cavity-block phase that the current minimal selective library does not correct. By contrast, the ECD-like library again performs poorly in the chosen encoding.

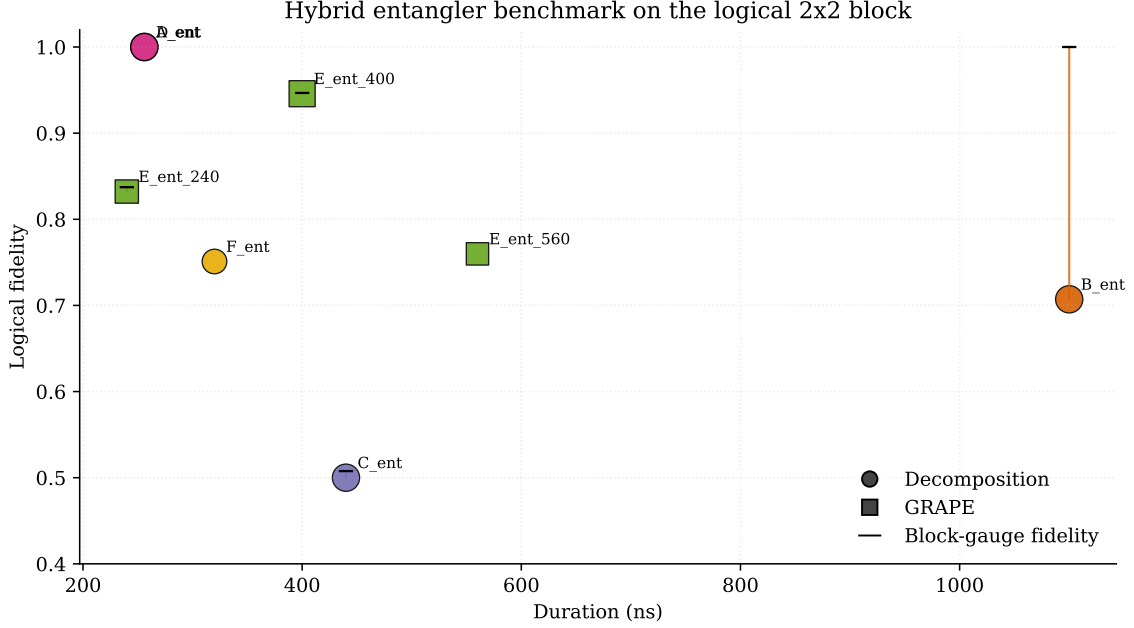


Figure 2: Entangler benchmark for $CX(c \rightarrow q)$. The native dispersive wait plus qubit rotations dominates the Pareto front. Gate Set B achieves perfect block-gauge fidelity, revealing a missing local phase correction rather than a lack of entangling structure.

Table 3: Best entangler candidates for the hybrid $CX(c \rightarrow q)$ target.

Library	Duration (ns)	F_{strict}	F_{block}	Leakage	Score J
D: native chi-wait + R_q	256	1.0000	1.0000	0.0000	0.9949
A: same entangler inside baseline library	256	1.0000	1.0000	0.0000	0.9949
E: GRAPE (400 ns)	400	0.9458	0.9467	0.0854	0.9207
F: native SWAP/sideband	320	0.7510	0.7510	0.1834	0.7079
B: single SQR	1100	0.7071	1.0000	0.0000	0.6851
C: ECD-like CD	440	0.5000	0.5077	0.0000	0.4912

3.3 Goal 3: Native vs Selective Control Routes

The dedicated native-vs-selective comparison is shown in Figure 3. For the local cavity task, the best selective route is still Gate Set A, the best native route is Gate Set F, and the best GRAPE pulse lands between them: GRAPE is faster, the native SWAP/sideband route is more interpretable, and SNAP remains the accuracy leader. For the entangler, the picture changes completely. The decisive resource is native control: once the chi-wait entangler is admitted explicitly, native control dominates both the selective SQR library and the GRAPE reference on fidelity, leakage, and duration. In that sense, the enlarged study sharpens the original conclusion. The practical winner is not “selective” or “native” in the abstract, but a split answer: selective dispersive control for the local cavity gate, native interactions for the entangler.

The new Gate Set F also adds an important nuance. Native exchange interactions are not merely future work; they already outperform the slower selective SQR library on the present

benchmark. What stops them from winning is not lack of hybridity or speed, but logical exactness and leakage in the strict $\{|0\rangle, |1\rangle\}$ encoding. This is exactly the sort of regime where a later cat/binomial study could plausibly reverse the ranking.

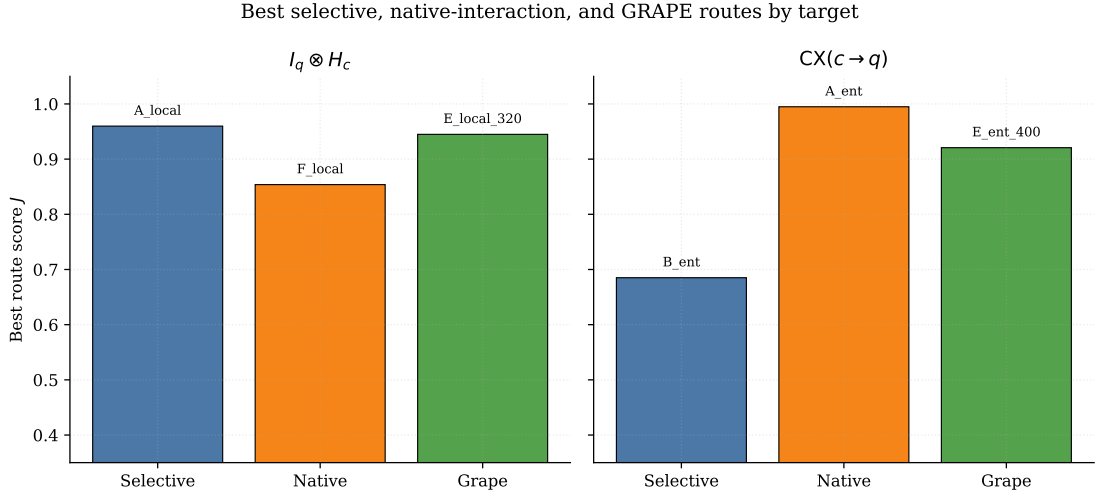


Figure 3: Best selective, native, and GRAPE routes for each target. Selective control wins the local cavity gate through D -SNAP- D , whereas native control wins the entangler through the chi-wait primitive. The SWAP-/sideband family is the strongest non-chi native local competitor but remains leakier than the selective baseline.

4 Validation

4.1 Sanity Checks

The expanded study outputs satisfy several direct sanity checks. First, the native dispersive entangler reproduces the target $CX(c \rightarrow q)$ to machine precision on the logical block, both in the isolated native library D and in the baseline library A when that same primitive is used. Second, the SQR entangler reaches block-gauge fidelity 1.0, confirming that the poor strict fidelity originates in a residual logical-phase mismatch rather than in missing entangling structure. Third, the new SWAP-/sideband-style library beats the selective SQR library on the local task and on strict entangler fidelity, confirming that it is a meaningful native benchmark rather than a contrived weak ansatz. Finally, the ECD-like library consistently underperforms in the strict Fock encoding, which agrees with the observed large leakage for the local cavity benchmark.

4.2 Convergence Analysis

The winning decomposition results are stable against cavity truncation. The local D -SNAP- D sequence was rerun at $n_{\text{cav}} = 6, 8, 10$, and the strict fidelity stayed within 1×10^{-7} of 0.988734. The native entangler is even cleaner: its strict fidelity remains 0.9999999263 at all three truncations with zero measured logical leakage. Higher-budget reruns of both winning decomposition ansatzes returned exactly the same nominal fidelities, so the decomposition conclusions are not limited by optimizer noise.

The GRAPE reference required a more careful convergence interpretation. Coarse single-seed grids were not reliable enough, so validation was performed in a best-of-seed fine-grid regime at fixed duration. For the local benchmark at 480 ns, the best strict fidelities over three seeds were 0.9713, 0.9638, and 0.9843 for 20, 24, and 32 time slices. For the entangler at 400 ns, the corresponding values were 0.9737, 0.9397, and 0.9317 for 20, 24, and 28 slices. These variations

are larger than for the structured decompositions, but they are small enough to support the qualitative benchmark conclusion that GRAPE does not beat the native entangler and only challenges the SNAP route on local-gate duration.

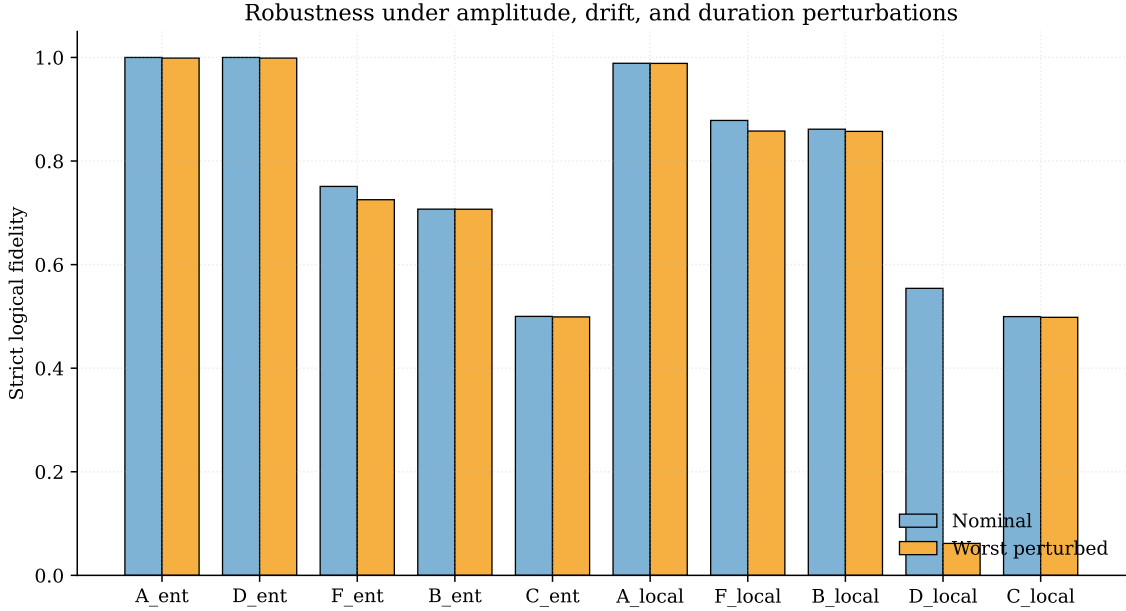


Figure 4: Worst-case strict fidelity under $\pm 2\%$ amplitude scaling, $\pm 1\%$ dispersive-shift scaling, and $+2\%$ duration stretch for the decomposition libraries. The winning Gate Set A local gate and the native chi-wait entangler remain the most stable candidates.

4.3 Literature Comparison

The observed hierarchy is consistent with the literature. The SNAP-based route is indeed accurate and universal but slow because it depends on selective operations [3, 4]. The ECD-like route is not disproven by this study; rather, the present results suggest that its main advantages are likely to emerge in encodings more naturally aligned with displacement geometry, as emphasized by the weak-dispersive ECD and CNOD literature [6, 7]. The sideband/exchange literature also matches the present native-gate outcome: these interactions are naturally hybrid and can be very fast [8–10], but exact computational-basis matching is not automatic. That last point is an inference from combining the literature with the present numerics: the papers motivate exchange as a serious fast primitive, while the benchmark shows that logical gauge control and leakage remain the main obstacles in the smallest Fock block. Finally, the native-interaction perspective from Ref. [11] matches the decisive entangler result: when the hardware already supplies a useful nonlinear interaction, that native route can dominate more elaborate selective or numerically optimized constructions.

5 Discussion

The central answer to the study question is that the best practical route is a hybrid strategy that combines selective and native ideas rather than betting on a single control family. For entanglement, the local device model already provides the right primitive: wait under dispersive evolution for $\pi/|\chi|$, and dress that phase with fast qubit rotations. For local cavity control, however, the older SNAP-plus-displacement construction still wins because it gives exact, compact, and robust logical action in the chosen Fock block. Taken together, those two facts point to an

augmented dispersive library rather than to a pure ECD route, a pure SWAP route, a pure SQR route, or a pure GRAPE route.

The enlarged native comparison also clarifies what the fast alternatives are actually good for. The SWAP-/sideband-style library is not the winner here, but it does beat the slower selective SQR library on both local-control score and strict entangler fidelity. That matters because it shows the native-exchange path is already competitive in the simulator, not just attractive in principle. What holds it back in this particular benchmark is leakage and basis matching, not lack of hybrid expressivity. By contrast, the ECD-like library remains poorly matched to the strict $\{|0\rangle, |1\rangle\}$ test even after the study-local wrapper was replaced by a first-class synthesis primitive.

GRAPE remains valuable as a benchmark and as a possible source of compressed pulse motifs, but its calibration burden and residual optimization sensitivity make it less attractive as the primary answer when the native dispersive entangler is already nearly perfect. The most important caveat is scaling. The present ranking is specific to the smallest hybrid logical block. As the cavity logical dimension grows, the cost of SNAP-based local control is expected to rise, and this is exactly the regime in which ECD-, CNOD-, or exchange-inspired control may become superior. The current recommendation should therefore be read as the right answer for the 2x2 Fock benchmark, not as a universal answer for all bosonic encodings.

6 Conclusion

The study goals were met. The local `cqed_sim` codebase was audited for hybrid-control capability, a literature-backed design memo was produced, the synthesis layer was extended with first-class native primitives for conditional displacement and SWAP-/sideband-style exchange, a reusable 2x2 logical benchmark was implemented, multiple gate libraries were compared against common targets, a GRAPE reference study was run, and the results were validated against truncation and solver-budget checks.

For the present device model and logical encoding, the best practical route to hybrid universal control is an augmented dispersive library:

$$\{R_q, D, \text{SNAP}, \text{native } \chi\text{-wait}\}. \quad (16)$$

The native dispersive conditional phase gives the best entangler by a wide margin, while D -SNAP- D remains the best local cavity basis-change tool in the strict 2x2 Fock benchmark. Native SWAP-/sideband-style control is now a serious explicitly benchmarked candidate, and its performance suggests that it should be revisited first when the study expands to larger or less Fock-like encodings. GRAPE remains useful as a pulse benchmark and a possible compression tool, but not as the preferred primary route in this first enlarged study.

7 Limitations and Future Work

7.1 Known Limitations

The first limitation is encoding scope. All conclusions were drawn for the Fock-encoded cavity pair $\{|0\rangle, |1\rangle\}$. This favors exact low-photon-number phase control and may underestimate the value of displacement-based fast control or exchange-style native control for cat-like or other non-Fock logical bases.

The second limitation is the native implementation path. Although the study now uses first-class `ConditionalDisplacement`, `JaynesCummingsExchange`, and `BlueSidebandExchange` primitives, these are still ideal effective-interaction models in the synthesis layer. They are not yet backed by a waveform bridge or a calibrated pulse-export workflow, so the native results

should be read as simulator-level gate-set benchmarks rather than as fully compiled experimental schedules.

The third limitation is solver maturity on the GRAPE side. Although the GRAPE benchmark was validated in a fine-grid best-of-seed regime, it still showed more optimization sensitivity than the structured decompositions. The GRAPE numbers therefore serve as a useful lower-bound benchmark, but not yet as a fully converged optimum.

The fourth limitation is ancilla modeling. The decomposition study used a two-level ancilla for tractability, which omits direct $|f\rangle$ -manifold leakage in the synthesis loop. That omission does not affect the main logical comparison qualitatively, but it does limit how literally the present fidelities should be interpreted experimentally, especially for sideband/exchange families.

7.2 Suggested Improvements

- **[P1 | HIGH]** Add waveform-bridge / pulse-export support for `ConditionalDisplacement`, `JaynesCummingsExchange`, and `BlueSidebandExchange`. This is the main remaining framework gap between the present synthesis benchmark and full simulator-level replay.
- **[P1 | MEDIUM]** Repeat the entire benchmark on a cat-like two-dimensional cavity encoding. The present Fock benchmark likely disadvantages displacement-native and exchange-native control families.
- **[P2 | MEDIUM]** Upgrade the winning libraries to $n_{\text{tr}} = 3$ noisy replay so direct $|f\rangle$ -level leakage can be quantified.
- **[P2 | MEDIUM]** Add a local cavity block-phase correction primitive to complement SQR. The present single-SQR entangler achieved perfect block-gauge fidelity and may become an exact high-quality selective library once the phase mismatch is corrected cheaply.
- **[P2 | MEDIUM]** Add warm-started and uncertainty-aware GRAPE to reduce seed sensitivity and make the pulse benchmark more comparable to the structured native libraries.
- **[P3 | MEDIUM]** Extend the target set beyond the Hadamard and a single CX to include random logical $\text{SU}(4)$ samples or a small hybrid Clifford benchmark.

7.3 Open Questions

The most important open question is whether the present winner remains dominant once the cavity logical dimension grows. The study strongly suggests that the local cavity-control bottleneck, not the entangler, is where the ranking could change. A second open question is whether the SWAP-/sideband-style native family would surpass the SNAP local gate in cat-like or binomial encodings, where exchange structure may be more naturally aligned with the logical basis. A third open question is whether an ECD- or CNOD-inspired local cavity primitive can replace SNAP while preserving the exact native entangler. Finally, it remains to be tested whether the best GRAPE pulses can be compressed into a short interpretable library that is easier to calibrate experimentally.

References

- [1] Alexandre Blais, Ren-Shou Huang, Andreas Wallraff, S. M. Girvin, and Robert J. Schoelkopf. Cavity quantum electrodynamics for superconducting electrical circuits: An architecture for quantum computation. *Physical Review A*, 69:062320, 2004. doi: 10.1103/PhysRevA.69.062320.

- [2] Jens Koch, Terri M. Yu, Jay Gambetta, A. A. Houck, D. I. Schuster, J. Majer, Alexandre Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf. Charge-insensitive qubit design derived from the Cooper pair box. *Physical Review A*, 76:042319, 2007. doi: 10.1103/PhysRevA.76.042319.
- [3] Reinier W. Heeres, Brian Vlastakis, Eric Holland, Stefan Krastanov, Victor V. Albert, Luigi Frunzio, Liang Jiang, and Robert J. Schoelkopf. Cavity state manipulation using photon-number selective phase gates. *Physical Review Letters*, 115:137002, 2015. doi: 10.1103/PhysRevLett.115.137002.
- [4] Stefan Krastanov, Victor V. Albert, Chao Shen, Chang-Ling Zou, Reinier W. Heeres, Brian Vlastakis, Robert J. Schoelkopf, and Liang Jiang. Universal control of an oscillator with dispersive coupling to a qubit. *Physical Review A*, 92:040303, 2015. doi: 10.1103/PhysRevA.92.040303.
- [5] Reinier W. Heeres, Philip Reinhold, Nissim Ofek, Luigi Frunzio, Liang Jiang, Michel H. Devoret, and Robert J. Schoelkopf. Implementing a universal gate set on a logical qubit encoded in an oscillator. *Nature Communications*, 8:94, 2017. doi: 10.1038/s41467-017-00045-1.
- [6] Alec Eickbusch, Volodymyr Sivak, Andy Z. Ding, Salvatore S. Elder, Shantanu R. Jha, Jayameenakshi Venkatraman, Baptiste Royer, S. M. Girvin, Robert J. Schoelkopf, and Michel H. Devoret. Fast universal control of an oscillator with weak dispersive coupling to a qubit. *Nature Physics*, 18:1464–1469, 2022. doi: 10.1038/s41567-022-01776-9.
- [7] Asaf A. Diringer, Eliya Blumenthal, Avishay Grinberg, Liang Jiang, and Shay Hacohen-Gourgy. Conditional-not displacement: Fast multioscillator control with a single qubit. *Physical Review X*, 14:011055, 2024. doi: 10.1103/PhysRevX.14.011055.
- [8] A. Wallraff, D. I. Schuster, A. Blais, J. M. Gambetta, J. Schreier, L. Frunzio, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf. Sideband transitions and two-tone spectroscopy of a superconducting qubit strongly coupled to an on-chip cavity. *Physical Review Letters*, 99:050501, 2007. doi: 10.1103/PhysRevLett.99.050501.
- [9] Fernando Valadares, Ni-Ni Huang, Kyle Timothy Ng Chu, Aleksandr Dorogov, Weipin Chua, Lingda Kong, Pengtao Song, and Yvonne Y. Gao. On-demand transposition across light-matter interaction regimes in bosonic cqed. *Nature Communications*, 15:5816, 2024. doi: 10.1038/s41467-024-50201-7.
- [10] Jordan Huang, Thomas J. DiNapoli, Gavin Rockwood, Ming Yuan, Prathyankara Narasimhan, Eesh Gupta, Mustafa Bal, Francesco Crisa, Sabrina Garattoni, Yao Lu, Liang Jiang, and Srivatsan Chakram. Fast sideband control of a weakly coupled multimode bosonic memory, 2025. URL <https://arxiv.org/abs/2503.10623>.
- [11] Axel M. Eriksson, Theo Sepulcre, Mikael Kervinen, Timo Hillmann, Marina Kudra, Simon Dupouy, Yong Lu, Maryam Khanahmadi, Jiaying Yang, Claudia Castillo-Moreno, Per Delsing, and Simone Gasparinetti. Universal control of a bosonic mode via drive-activated native cubic interactions. *Nature Communications*, 15:2512, 2024. doi: 10.1038/s41467-024-46507-1.
- [12] SoraUmika and contributors. qubox_cqedsim: circuit-QED simulation framework, 2025. URL https://github.com/SoraUmika/qubox_cQEDsim.